

Shreyansh Agarwal

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EDUCATION

LNMIIT Institute of Information Technology, Jaipur

B.Tech in Computer Science Engineering

Aug 2023 – May 2027

CGPA: 7.17

EXPERIENCE

Eficaz HR Services Private Limited

AI Engineer Intern

May 2026 – July 2026

Remote

- Developed AI-powered applications using **LLMs, LangChain and Retrieval-Augmented Generation (RAG)**, improving response relevance for context-aware user queries.
- Built document intelligence pipelines using **vector embeddings** and semantic search across vector databases, cutting manual information-retrieval time significantly.
- Integrated LLM APIs with backend services and optimized prompt engineering workflows to improve AI response quality and reliability.

PROJECTS

ScaleCart – High-Scale E-Commerce Backend | GitHub | Java, Spring Boot, PostgreSQL, Redis, JPA/Hibernate

- Engineered a full-fledged e-commerce backend with **25+ RESTful API endpoints** across product, inventory, cart, order and user modules using Spring Boot and PostgreSQL.
- Implemented transaction-safe order processing with database transactions and concurrency control (optimistic locking), eliminating overselling across **50+ simulated concurrent requests**.
- Integrated Redis caching for high-traffic product and catalog endpoints, cutting average API response time by **~60%** (250ms → 100ms) and reducing direct database load by **~70%**.
- Applied database indexing, optimized JPQL queries and JPA/Hibernate entity mapping, improving read-query throughput by **~3x** while preserving consistency across order and inventory operations.

Truth Engine – Multi-Source RAG & Contradiction Resolution System | GitHub | Python, FastAPI, FAISS, BM25

- Architected a hybrid retrieval engine blending FAISS dense search with BM25 sparse search across 3 independent sources, fusing top-10 candidates per algorithm via **Reciprocal Rank Fusion (k=60)**.
- Developed a multi-format ingestion pipeline (PDF, JSON, CSV, Markdown, TXT, up to 20MB) using sliding-window chunking (512-token chunks, 50-token overlap) and **384-dim embeddings** batched at size 64.
- Implemented cross-encoder re-ranking (ms-marco-MiniLM-L-6-v2, ~10ms/chunk) with sigmoid-normalized confidence scoring across High/Medium/Low tiers, plus contradiction detection across **4 severity levels**.
- Engineered a zero-downtime 3-stage LLM fallback chain (Gemini Flash → Groq LLaMA-3.3-70B → local Ollama) with 15s timeouts, supporting **10 concurrent sessions** with automatic retry on HTTP 429/5xx, validated with Pytest.

AI Trip Planner | GitHub | Python, LangGraph, LangChain, FastAPI, Streamlit

- Architected an agentic AI travel-planning system using LangGraph StateGraph with **6+ modular agent nodes** to orchestrate multi-step reasoning and tool-calling with Groq-hosted LLMs.
- Integrated **4 external APIs** (OpenWeatherMap, Google Places, Tavily Search, currency-conversion) to auto-generate real-time, context-aware, multi-day itineraries in under 10 seconds.
- Designed a FastAPI backend exposing **10+ REST endpoints** to serve trip-planning logic to a Streamlit-based conversational interface.
- Implemented modular agent nodes for state management, tool routing and response synthesis, enabling scalable workflow orchestration across multi-city trip requests.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL

Backend & APIs: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, REST APIs, FastAPI, Flask

Databases: PostgreSQL, MySQL, MongoDB, Redis, FAISS, ChromaDB

Developer Tools: Git, GitHub, Docker, GitHub Actions, CI/CD, Linux, Postman

AI/ML: Machine Learning, NLP, Generative AI, LLMs, RAG, LangChain, LangGraph, BM25

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

Solved **500+** Data Structures and Algorithms problems across LeetCode and other coding platforms.

Coordinator, Basketball – LNMIIT: Managed sports activities and tournaments.

Logistics Head, Vivacity: Led a team for college fest operations.